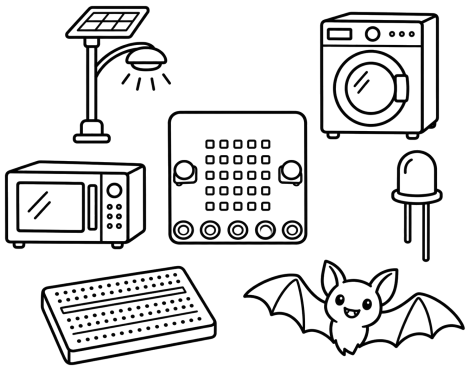
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 9



UNIT 4
Use of Microcontrollers

Student Workbook

"The King of ICT"

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Motivational Speaker
BSc (Sp) Computer Science - University of Peradeniya
PGDE - Open University of Sri Lanka
MIT (Reading) - University of Sabaragamuwa

STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain how a sensor and a microcontroller work as input, process and output.
- [ ] Name the four basic components of a microcontroller.
- [ ] Name devices that use microcontrollers, like the solar lamp.
- [ ] Label the front and the back of the micro:bit and connect it to a computer.
- [ ] Code the micro:bit with the blocks of the Basic and Input menus.
- [ ] Label the Arduino Uno board, and use a breadboard and an LED.
- [ ] Explain how the PIR sensor and the ultrasonic sensor work.

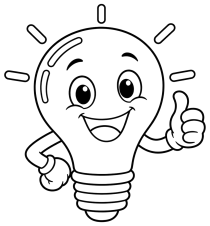
THREE RULES FOR THE COMPUTER ROOM

- **Always** ask your teacher before you connect a board or a sensor.
- **Never** connect a circuit to the wall socket (mains electricity).
- **Always** save your micro:bit project with your name.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task with a micro:bit or an Arduino. Last you **write** the exercises.

Colour the small pictures when you finish your work early.



Colour me!

THE WORDS YOU WILL LEARN

Sensor	Microcontroller	Input	Process
Output	CPU	Non-volatile memory	Volatile memory
System clock	Peripheral devices	SBC / SCC	Bluetooth
micro:bit	Block editor	on start	forever
Arduino Uno	ATMEGA328p	Breadboard	LED
PIR sensor	Ultrasonic sensor	VCC / GND	Echo

"A tiny chip runs our lamps, ovens and washing machines. Learn how it thinks!"

- THE KING OF ICT

CHAPTER 1

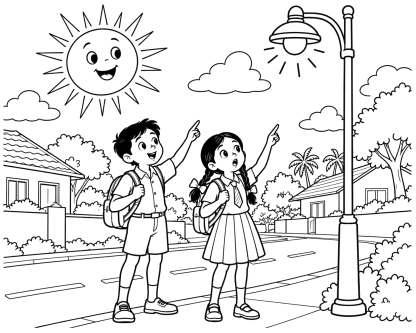
Sensors and Microcontrollers

? WHO SWITCHES THE STREET LIGHTS?

"The sun is up, but the street lights are still on!"

"A light sensor can switch them on and off."

Electricity is wasted. A **sensor** collects the data and a **microcontroller** decides what to do.



4.1 Sensors around us

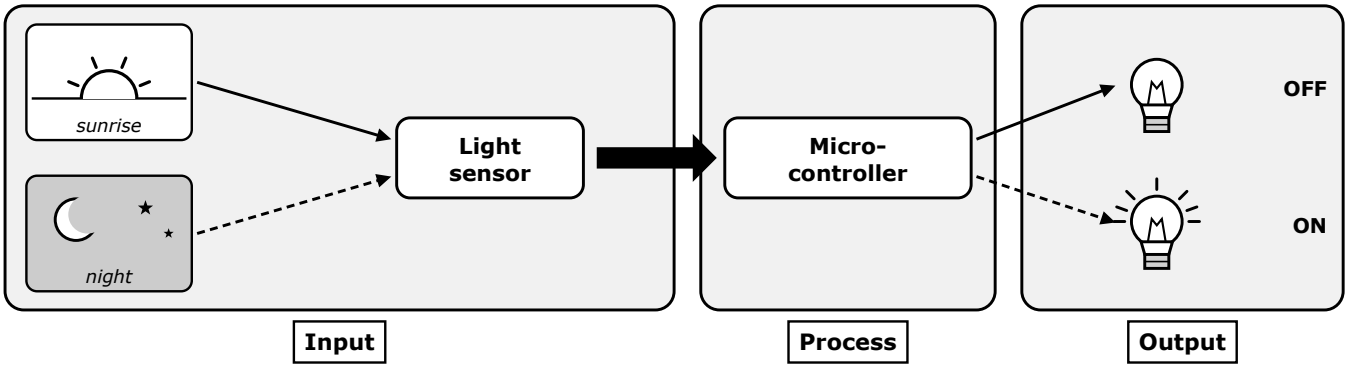
There are **sensors** to measure things like **soil conditions**, **temperature** and **environmental conditions**. Specialized computer programs are used to control them. A **light sensor** can detect sunrise and sunset. To use sensors to detect the conditions around us, they must be **connected to a computer**.

4.2 Input, process and output

The basic functions of a computer are to **input** (collect) data, to **process** the data according to the given instructions, and to **output** the necessary information.

Input or collect data → **Process data** → **Output necessary information**

A **microcontroller** processes the data collected from sensors according to the given instructions, and releases the necessary output. Here a light sensor and a microcontroller light a bulb or put it off:



An example of microcontroller operation (textbook p. 48). Solid arrows: sunrise, the bulb is OFF. Dashed arrows: night, the bulb is ON.

For this, the microcontroller needs to be fed with **program instructions** in a program.

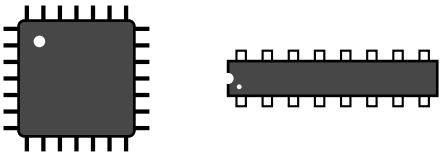
"A smart sensor saves electricity. A smart student saves time."

- THE KING OF ICT

4.3 What is a microcontroller?

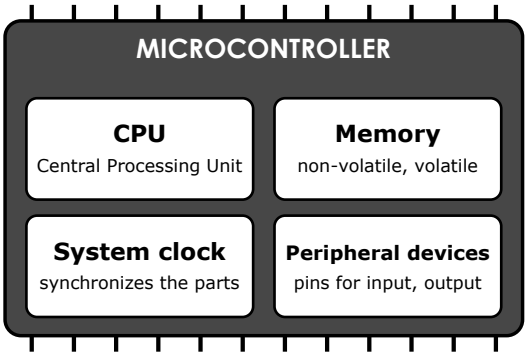
A **microcontroller** is a **chip** designed to perform some operations according to the given instructions.

Microcontrollers work in many devices, like solar lamps, washing machines and microwave ovens (4.5).



Two microcontroller chips

4.4 The four components of a microcontroller



- **CPU** (Central Processing Unit): it processes data into information, to give the output.
- **Memory**: non-volatile memory and volatile memory (see below).
- **System clock**: an electronic device that **synchronizes** all the components.
- **Peripheral devices**: small **pins** used to input data or output information. Data is collected in **analog** and **digital** forms; the output is released in **digital** form.

(i) NON-VOLATILE MEMORY

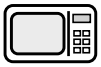
Its data is not erased even if there is no electricity. The program that the microcontroller should execute is stored in it.

(i) VOLATILE MEMORY (RAM)

Its data is erased when there is no electricity. It works as the **RAM** (Random Access Memory): it stores data and results while the microcontroller is at work, and the data and instructions for the CPU.

4.5 Microcontrollers in daily life

Many devices use sensors and microcontrollers (textbook p. 50):

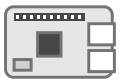
	Device	How it uses the microcontroller
	**Solar lamp**	It has **solar cells**, **light sensors** and a **microcontroller**. They work together: the lamp turns **on** automatically when it is dark and turns **off** when there is light. Solar lamps minimize the waste of electricity.
	**Washing machine**	The user **presses buttons** to give instructions to the microcontroller for the wash. The microcontroller operates accordingly.
	**Microwave oven**	The user gives the required **temperature** and the **duration**. The microcontroller holds the heat for that period and then stops.



Traffic lights, **computer printers** and **remote controllers** also use microcontrollers (textbook summary, p. 65).

4.6 Single board and single chip computers

SINGLE BOARD COMPUTER (SBC)



One single circuit board with **memory**, **input**, **output** and a **microcontroller**. Example: **Raspberry pi**.

SINGLE CHIP COMPUTER (SCC)



The **CPU**, **input**, **output** and **memory** are all built into a single integrated circuit (**IC**). Example: the **Arduino chip**.

DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 4.1)

Devices with sensors and microcontrollers. Work in pairs. **Ask your teacher first.**

1	Look around your home and your school for devices that work with sensors or buttons.
2	Find devices from the textbook: a solar lamp, a washing machine, a microwave oven, a printer ...
3	For each device, find the **input**: what does its sensor detect, or which buttons are pressed?
4	Find the **output**: what does the device do? Remember: the microcontroller does the **process**.
5	Fill the table below for three devices. Tell the class which device saves electricity.

Device	Input	Output

A Match the components

6 marks

Write the correct letter in the box.

A - Name		B - What it is
1. CPU		**A.** An electronic device that synchronizes all the components.
2. Non-volatile memory		**B.** A chip designed to perform operations according to the given instructions.
3. Volatile memory		**C.** Small pins used to input data or output information.
4. System clock		**D.** Its data is erased without electricity. It works as the RAM.
5. Peripheral devices		**E.** It processes data into information to give the output.
6. Microcontroller		**F.** Its data is not erased without electricity. The program is stored in it.

B Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

process	sensor	chip	non-volatile
RAM	clock	digital	dark

1. The basic functions of a computer are input, ... and output.
2. A light ..... can detect sunrise and sunset.
3. A microcontroller is a ..... designed to perform operations according to instructions.
4. The program of a microcontroller is stored in the ..... memory.
5. Volatile memory functions as the ..... (Random Access Memory).
6. The system ..... synchronizes all the components.
7. The output of a microcontroller is released in ..... form.
8. A solar lamp turns on automatically when it is ..... .

C Inputs and outputs of devices

6 marks

Write the input and the output. The microcontroller does the process.

Device	Input	Output
Solar lamp		
Washing machine		
Microwave oven		

D True or false

8 marks

Write T or F in the box.

1.		Sensors can measure soil conditions and temperature.
2.		A microcontroller can work without any program instructions.
3.		Data in non-volatile memory is erased when there is no electricity.
4.		The data and instructions for the CPU are held in the volatile memory.
5.		A microcontroller has four basic components.
6.		The peripheral pins collect data only in digital form.
7.		A washing machine works according to the buttons the user presses.
8.		A Raspberry pi is a single chip computer.

E Input, process or output?

8 marks

Tick one box in each row.

Which part is it?	Input	Process	Output
The light sensor detects that it is getting dark.	☐	☐	☐
The microcontroller decides to switch on the street lamp.	☐	☐	☐
The street lamp lights up.	☐	☐	☐
The user presses the buttons of the washing machine.	☐	☐	☐
The microcontroller follows the program instructions.	☐	☐	☐
The microwave oven stops heating at the set time.	☐	☐	☐
The light sensor of a solar lamp detects sunlight.	☐	☐	☐
The CPU processes the data into information.	☐	☐	☐

F SBC or SCC?

4 marks

Tick one box in each row.

It is ...	Single board computer	Single chip computer
All the parts are on one circuit board.	☐	☐
The CPU, input, output and memory are in one IC.	☐	☐
A Raspberry pi	☐	☐
The Arduino chip	☐	☐

G Choose the correct answer

6 marks

Underline your answer.

1. A microcontroller consists of how many basic components?
   (a) two (b) three (c) four (d) five

2. Which component synchronizes all the components?
   (a) CPU (b) memory (c) system clock (d) pins

3. Where is the program of the microcontroller stored?
   (a) non-volatile memory (b) volatile memory
   (c) system clock (d) pins

4. Which device has solar cells, light sensors and a microcontroller?
   (a) washing machine (b) microwave oven
   (c) solar lamp (d) printer

5. The user of a microwave oven gives the required temperature and the
   (a) duration (b) colour (c) sensor (d) clock

6. In the light sensor example, what is the output?
   (a) the sunrise (b) the light sensor
   (c) the microcontroller (d) the bulb

CHAPTER 2

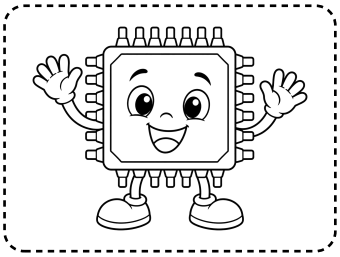
Microcontroller Kits and the micro:bit

? A COMPUTER IN YOUR POCKET?

"Is this tiny board really like a computer?"

"Yes! It has inputs, outputs and a memory."

Meet the **micro:bit**, a microcontroller based kit.



Colour me!

4.7 Microcontroller based kits and the micro:bit

A **microcontroller based kit** can be used to get the outputs we need. Three such kits are the **micro:bit**, the **Arduino** and the **Raspberry pi**. This unit is mostly about the micro:bit and the Arduino.

The **BBC** developed the **micro:bit** module to get **inputs**, **process** them and produce **outputs**. It contains a **memory** too, so the board has the basic features of a computer. It also contains **sensors**.

i WEBSITES

www.microbit.co.uk
www.arduino.cc
www.raspberrypi.org

4.8 The front and the back of the micro:bit

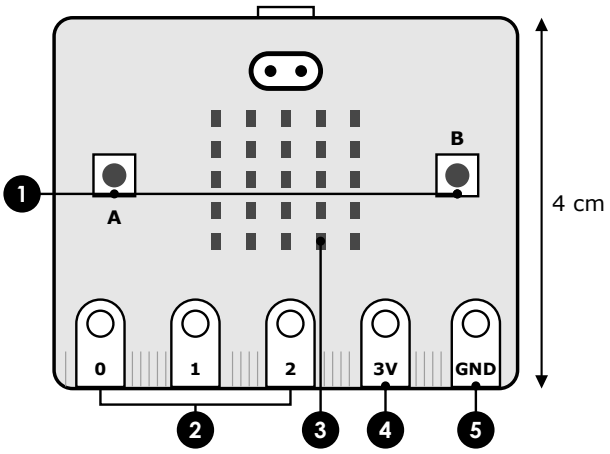


Figure 4.1 - the front view

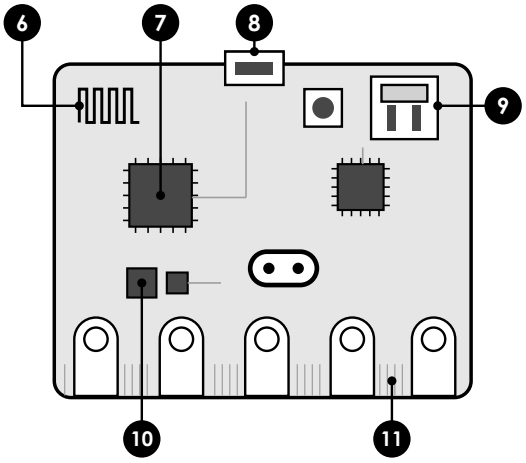


Figure 4.2 - the rear view

1. Two programmable buttons, A and B
2. Digital and analog input / output pins (0, 1 and 2)
3. LED bulbs (25) that can be programmed one by one. They are built into the module, so extra LED bulbs are not necessary.
4. Port to connect power (3V)
5. Ground back port (GND)
6. Bluetooth smart antenna: connects devices using Bluetooth and transmits radio waves
7. Central Processing Unit (CPU)
8. Micro USB port to connect to a computer
9. Battery connector for a 3 V external power supply
10. Accelerometer and compass (embedded sensors)
11. Pin edge connector

4.9 Connecting the micro:bit to a computer

The module is connected to the computer with a **micro USB cable**. When it is connected, the computer shows it as a **storage unit**.

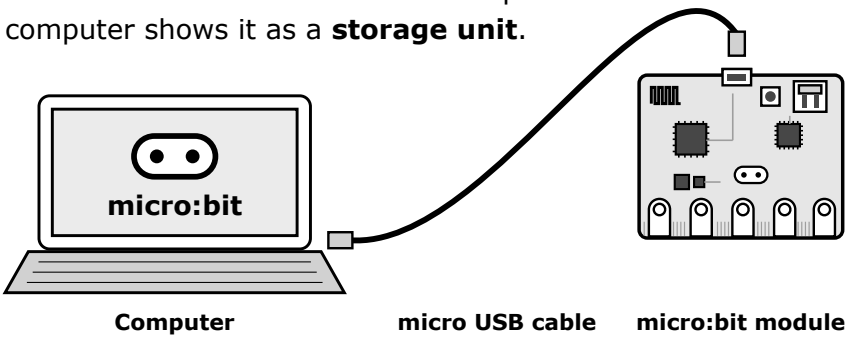


Figure 4.3 - connecting the micro:bit module

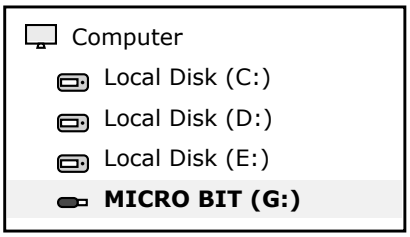


Figure 4.4 - the micro:bit as a storage unit

A Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

BBC 25 USB storage 3 embedded B kits

1. The micro:bit module was developed by the ..... .
2. The front of the micro:bit has ..... LED bulbs.
3. The micro:bit is connected to a computer with a micro ..... cable.
4. The computer shows the micro:bit as a ..... unit.
5. The battery connector is for a ..... V external power supply.
6. The accelerometer and the compass are ..... sensors.
7. The two programmable buttons are A and ..... .
8. micro:bit, Arduino and Raspberry pi are microcontroller based ..... .

B Which part do you use?

8 marks

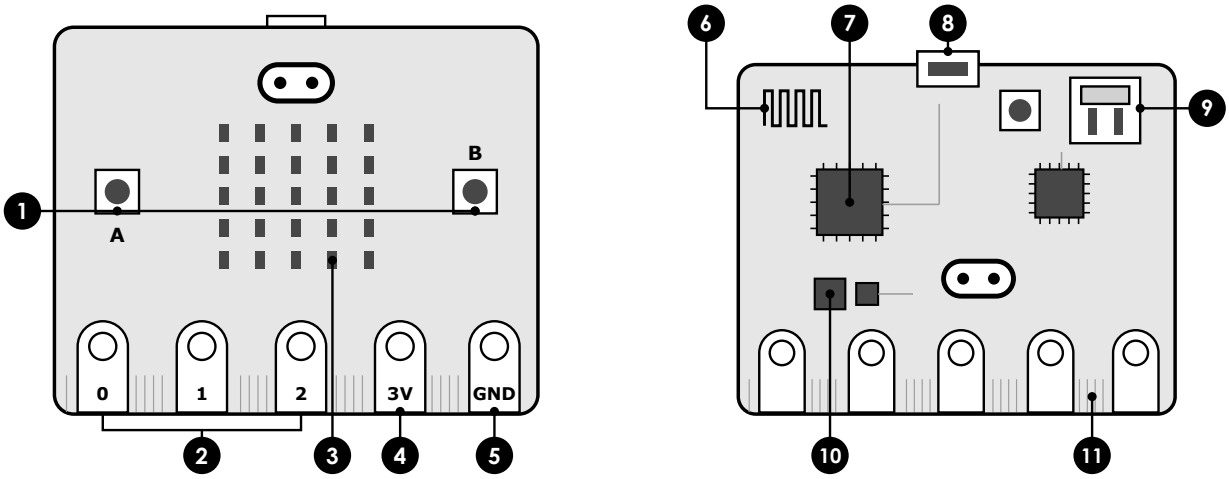
Write the part of the micro:bit.

You want ...	Part of the micro:bit
To connect the micro:bit to a computer	
To give an input by pressing it	
To show a pattern, a number or a word	
To connect a 3 V external power supply (batteries)	
To connect devices using Bluetooth and send radio waves	
To find the direction and the motion	
To process the data	
For digital and analog inputs and outputs	

C Label the micro:bit

11 marks

Write the name of each numbered part.



1. ..... 2. ..... 3. .....

4. ..... 5. ..... 6. .....

7. ..... 8. ..... 9. .....

10. ..... 11. .....

D True or false

6 marks

Write T or F in the box.

1.		The micro:bit has the basic features of a computer.
2.		Extra LED bulbs must be added to the micro:bit to show a pattern.
3.		The micro:bit can connect to other devices using Bluetooth.
4.		The CPU is on the front of the micro:bit.
5.		The micro:bit module is about 4 cm high.
6.		In Figure 4.4 the micro:bit is shown as Local Disk (C:).

CHAPTER 3

Coding the micro:bit with Blocks

? CAN A BEGINNER CODE IT?

"I have never written a program. Can I code it?"

"Yes! Just drag and drop the blocks."

The micro:bit is coded online in a **block editor**.



4.10 Coding the micro:bit online

The micro:bit is coded **online**, in the website **makecode** (www.makecode.com). Its **drag-and-drop** facility speeds up programming, so even a beginner can follow the process. It can be used in **Sinhala** too: **More** → **Language** → **Sinhala**.

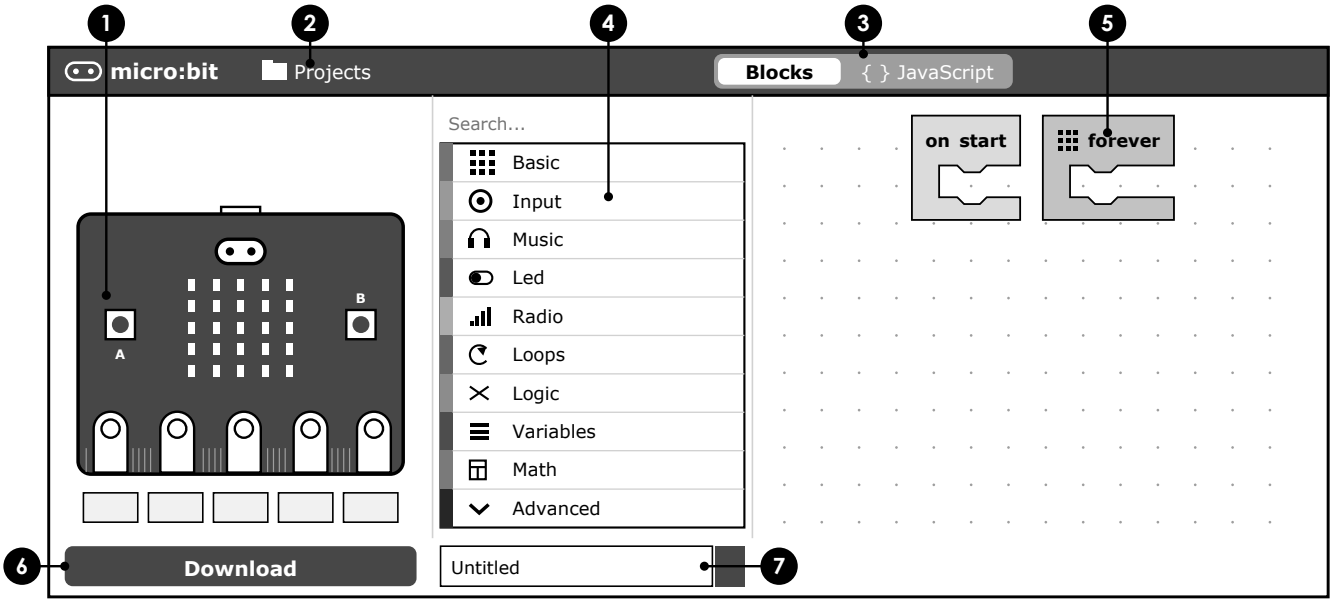


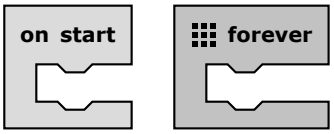
Figure 4.7 - starting a new project in micro:bit (the block editor)

1. the micro:bit model: shows how the program works
2. Projects (a new project starts here)
3. Blocks or JavaScript
4. the block editor menu
5. the on start and forever blocks
6. Download
7. the name of the project (Untitled)

4.11 Starting a new project

Open **makecode.microbit.org** and click **Projects** → **New Project**. Click **Blocks**: coding is done by connecting blocks. Languages like **JavaScript**, **Python** and **C++** can be used too.

A new project shows two blocks (Figure 4.8). Put blocks **inside** them.



on start: the initial action is put here.

forever: repetitive work is put here.

4.12 The block editor menu

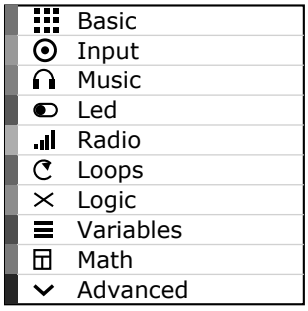
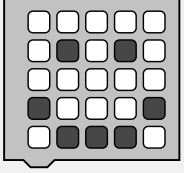


Figure 4.9 - the menu

To get more blocks, click a **menu** in the block editor. Each menu has its own colour, and its blocks have the same colour. The **Basic** and **Input** menus are below; the other menus are used in the activities.

Tip: once the coding is done, the program can be seen working on the micro:bit model in the block editor.

4.13 Blocks of the Basic menu

Block	What it does
show number 0	To display a number.
show leds 	To create different patterns using the 25 LEDs.
show icon	To select an LED pattern (an icon, like a heart).
show string "Hello!"	To display typing contents (words you type).
pause (ms) 100	To give an interval in milliseconds for an action (1000 ms = 1 second).
clear screen	To erase displayed data.
show arrow North	To display an arrow (North, South ...).

The Basic menu also has the **on start** and **forever** blocks (4.11).

4.14 Blocks of the Input menu

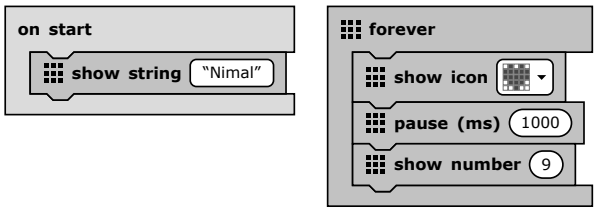
Block	What it does
on button A pressed	To specify actions for buttons A and B.
on shake	To specify actions for a shaking or a motion.
on pin P0 pressed	To specify actions for pins P0, P1 and P2.
button A is pressed	A six-sided block: is button A (or a pin) being pressed? It goes into other blocks.
acceleration (mg) x	To display accelerometer data.
light level	To display light sensor data.
compass heading (°)	Displays compass direction data.
temperature (°C)	Displays temperature sensor data.

Round blocks like **light level** go into the slot of another block, for example **show number**.

DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 4.2)

Your first program with the **Basic** blocks. **Ask your teacher first.**

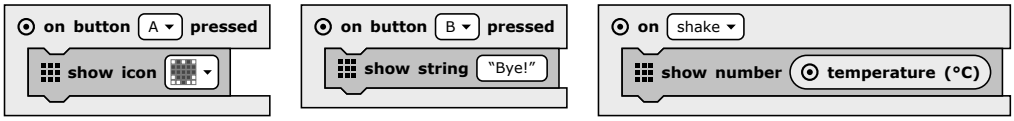
1	Open makecode.microbit.org. Click **Projects**, then **New Project**, then **Blocks**.
2	From the **Basic** menu, drag **show string** into **on start**. Type your name.
3	Into **forever**: **show icon** (a heart), **pause (ms)** 1000 and **show number** 9.
4	Watch your program on the micro:bit model on the left of the screen.
5	Connect the micro:bit (micro USB cable). Click **Download**; save the file on the **MICRO BIT** drive.



DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 4.3)

Use the **Input** blocks: buttons, shaking and sensors.

1	From the **Input** menu, drag **on button A pressed**. Put **show icon** in it.
2	Drag one more, change A to **B** and put **show string** "Bye!" in it.
3	Drag **on shake**; put **show number** in it and drop **temperature (°C)** into its slot.
4	Test it on the micro:bit model, then on the real micro:bit.
5	Try **show number** with **light level** in **forever**. Cover the micro:bit with your hand.

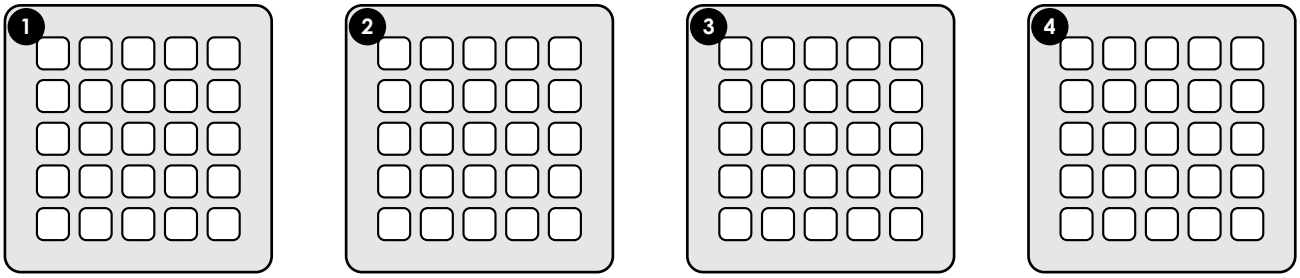


A Draw the LED patterns

8 marks

Shade the LEDs of each show leds block.

a plus sign **a big X** **a square on the edge** **an arrow (North)**



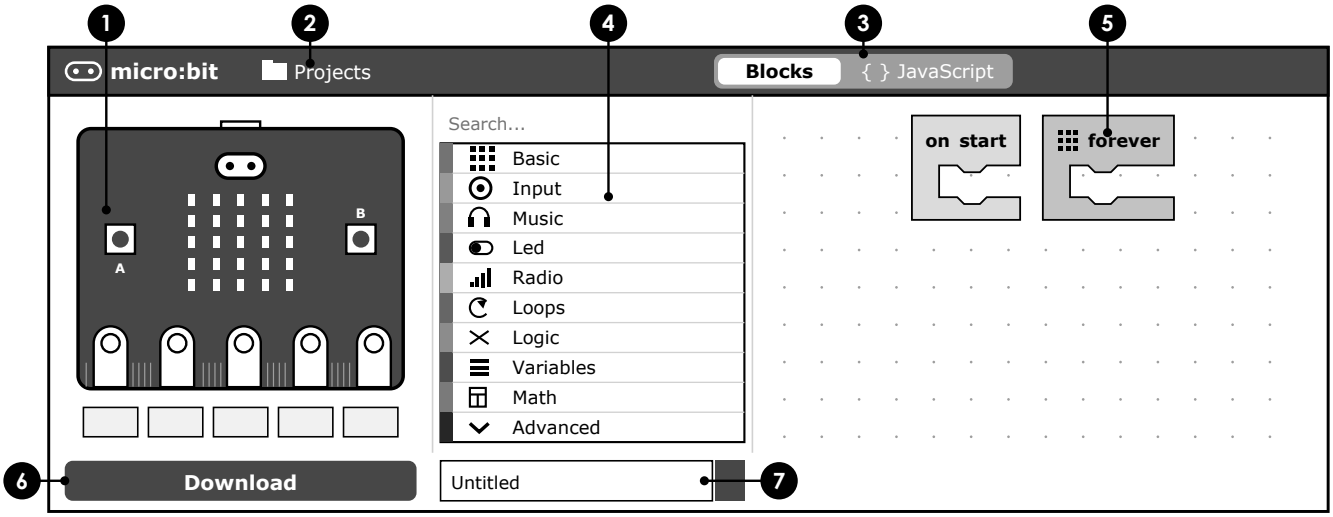
"Every great program starts with one block. Drag it, drop it, try it!"

- THE KING OF ICT

B Label the block editor

7 marks

Write the name of each numbered part.



1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. .....

C Basic or Input?

8 marks

In which menu is each block? Tick one box.

The block	Basic	Input
show number	☐	☐
on button A pressed	☐	☐
show leds	☐	☐
light level	☐	☐
pause (ms)	☐	☐
on shake	☐	☐
clear screen	☐	☐
temperature (°C)	☐	☐

D Answer these

4 marks

Write full sentences.

1. Why can even a beginner code the micro:bit? (2 marks)

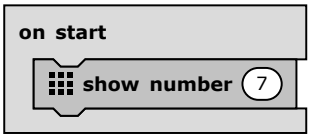
2. Name two other ways (languages) to code the micro:bit. (2 marks)

E What will the micro:bit show?

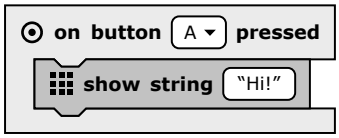
8 marks

Read each program. Write what happens.

1



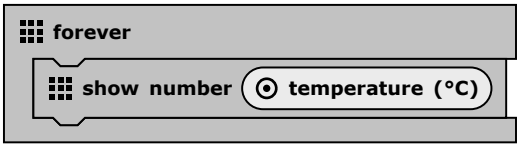
2



3



4



F Match the blocks

6 marks

Write the correct letter in the box.

A - Block		B - What it does
1. **show string**		**A.** Displays compass direction data.
2. **show leds**		**B.** To erase displayed data.
3. **pause (ms)**		**C.** To display typing contents.
4. **clear screen**		**D.** To specify actions for pins P0, P1 and P2.
5. **on pin P0 pressed**		**E.** To create different patterns using 25 LEDs.
6. **compass heading (°)**		**F.** To give an interval in milliseconds for an action.

"Do your best work on start, and keep practising forever."

- THE KING OF ICT

CHAPTER 4

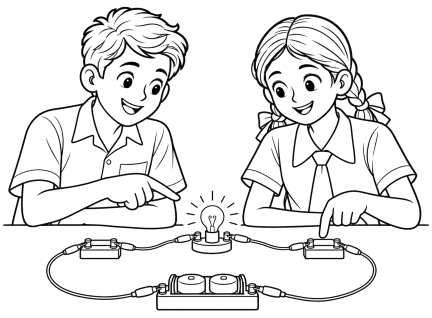
Arduino, Breadboards and Sensors

? BUILD IT ON A BOARD

"Can a board switch on a light when I come near?"

"Yes! Join a sensor and an LED to an Arduino."

Meet the **Arduino**, the **breadboard**, the **LED** and two **sensors**.



4.15 The Arduino board

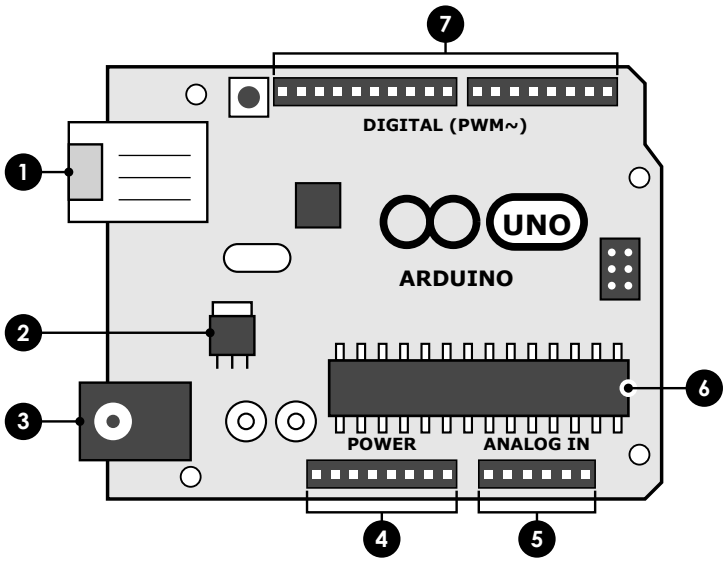


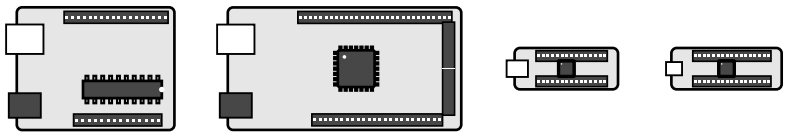
Figure 4.12 - the Arduino Uno board

The Arduino board has a **microcontroller** chip made by the **Atmel** company. It can receive **input**, **process** it and release **output**. It contains a **memory** as well.

So this board is similar to a **basic computer**.

No	Part	What it does (Table 4.1)
1	**Mini USB**	Used to connect the board to a computer.
2	**Power regulator**	Controls the voltage given to the Arduino board (Table 4.1 calls it the electric controller).
3	**DC power**	Connected to a computer, the board gets its 5 V from the computer. If not, this port supplies external power.
4	**Power pins**	Give electricity from the board to an external circuit. Also used to control some operations.
5	**Analog pins**	Send analog inputs (for example a sensor reading) to the board.
6	**ATMEGA328p**	The microcontroller chip of the Arduino Uno. It is produced by the Atmel company.
7	**Digital pins**	Get digital inputs and give digital outputs.

4.16 Arduino boards used today

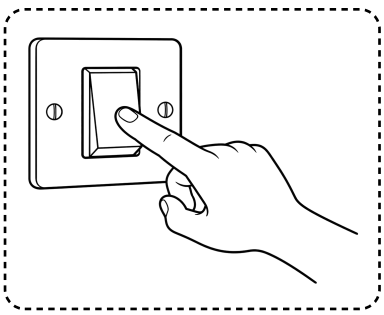


Uno Mega Nano Micro

A few Arduino boards used today: the **Uno**, the **Mega**, the **Nano** and the **Micro**. We use the **Arduino Uno** board for the activities in this unit.

"Listen like a bat, think like a microcontroller, and you will always find your way."

- THE KING OF ICT



Colour me!

4.17 The breadboard

Breadboards are used to make circuits without **welding** (soldering) the parts together.

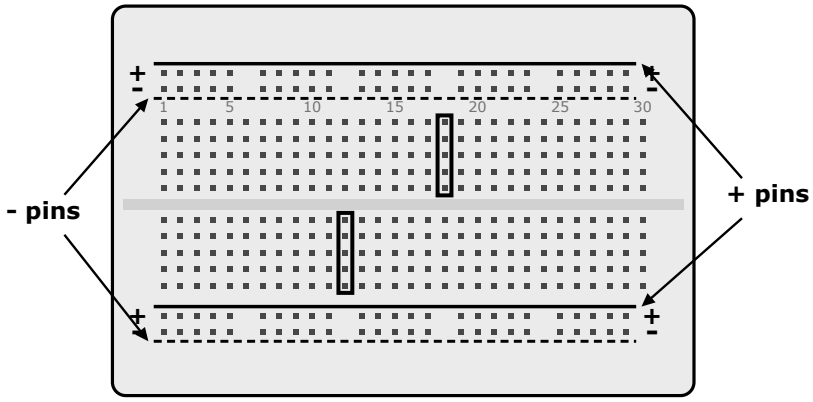


Figure 4.13 - a breadboard

On a breadboard:

- All the holes of a **(+)** row (the red line) are connected together.
- All the holes of a **(-)** row (the blue line) are connected together.
- The holes in each green section are connected together. In the drawing: each **column of five holes** (in a box) is connected, on one side of the middle groove only.

4.18 The Light Emitting Diode (LED)

A **Light Emitting Diode (LED)** emits light when electricity passes through it.

In a **diode**, electricity travels only in **one direction**. So an LED has a **+ pin** and a **- pin**, and they must be joined the right way.

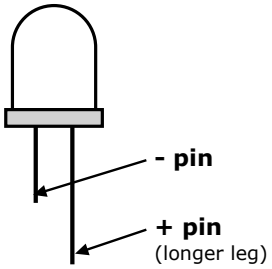


Figure 4.14 - an LED

4.19 Sensors and the PIR sensor

A **sensor** is a device which detects or measures a physical property and records, indicates or otherwise responds to it. The **Passive Infrared (PIR) sensor** measures infrared objects, like **humans**, in its field of view and gives an output voltage.

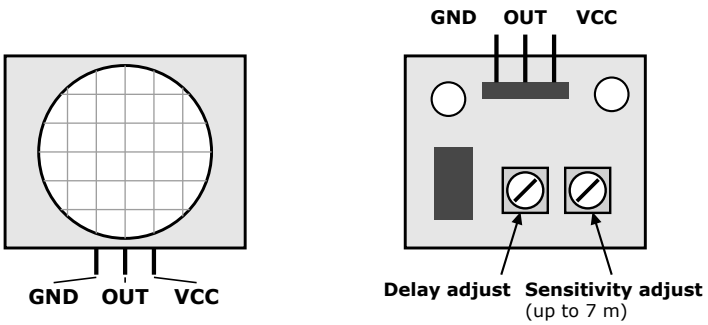


Figure 4.15 - the PIR sensor: front (left) and rear (right)

Pin	What it is
GND	the negative terminal
VCC	the positive terminal (needs 5 V)
OUT	the output terminal (outputs 3.3 V)

The two yellow **trimpots**: one adjusts the **distance** the sensor covers (up to 7 m), the other sets the **time delay** to release the output.

"Your kind words come back to you, like an echo."

- THE KING OF ICT

4.20 The ultrasonic sensor

An **ultrasonic sensor** estimates the **distance** from it to an object. It sends **ultrasound signals** to the object and interprets the **reflected signals** (Figure 4.16). It works like a **bat** finding its route (Figure 4.17).

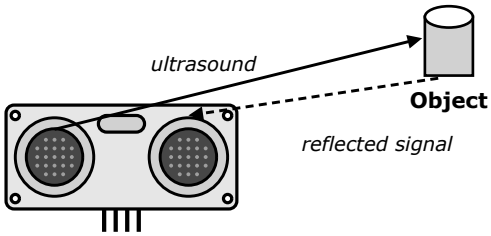


Figure 4.16 - how it works

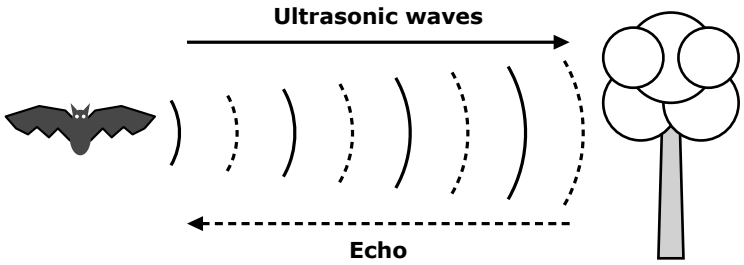


Figure 4.17 - a bat finds its route in the same way

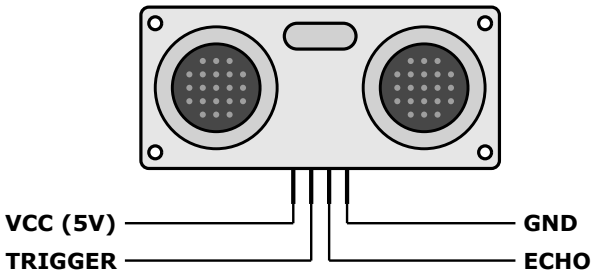


Figure 4.18 - the pins of the ultrasonic sensor

Pin	What it is
GND	the negative terminal
VCC	the positive terminal (5 V must be supplied to it)
TRIGGER	for input
ECHO	provides the output

DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 4.4)

Light an LED with the Arduino Uno. **Only your teacher connects the power.**

1	Find the seven parts of Figure 4.12 on your Arduino Uno board.
2	Push an LED into the breadboard. Find its **+ pin** (the longer leg) and its **- pin**.
3	With your teacher, join the + pin to a **digital pin** and the - pin to **GND** (a power pin).
4	Connect the board to the computer (it gets 5 V). Your teacher loads a program: watch the LED blink.

DO IT - PRACTICAL 5 (WORKBOOK ACTIVITY 4.5)

A PIR sensor finds a person. Use the circuit of Practical 4.

1	Find the three pins of the PIR sensor: **GND**, **OUT** and **VCC**.
2	With your teacher, join VCC to **5V**, GND to **GND** and OUT to a **digital pin**.
3	Your teacher loads the program. Walk in front of the sensor. Does the LED light up?
4	Turn the **sensitivity** and the **delay** trimpots a little. What changes?

DO IT - PRACTICAL 6 (WORKBOOK ACTIVITY 4.6)

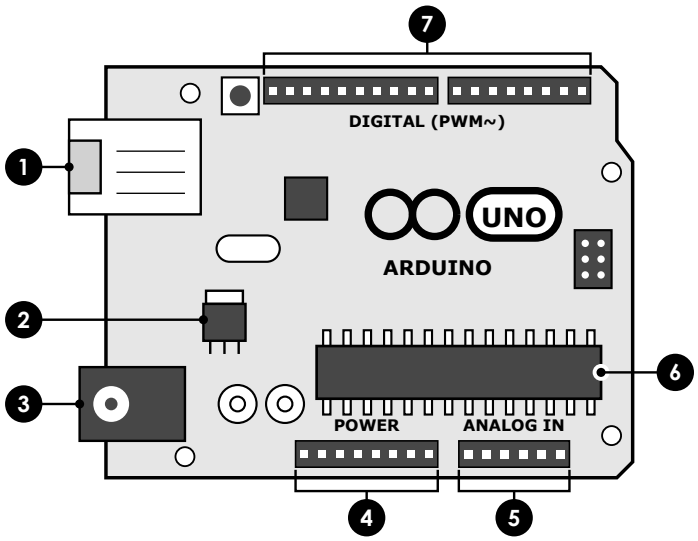
An ultrasonic sensor measures a distance. **Ask your teacher first.**

1	Find the four pins of the ultrasonic sensor: **VCC**, **TRIGGER**, **ECHO** and **GND**.
2	With your teacher, join VCC to **5V** and GND to **GND**, TRIGGER and ECHO to two digital pins.
3	Your teacher loads a program that shows the distance on the computer screen.
4	Move a book nearer, then farther: the distance changes. How is this like a bat?

A Label the Arduino board

7 marks

Write the name of each numbered part.

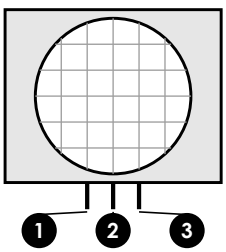


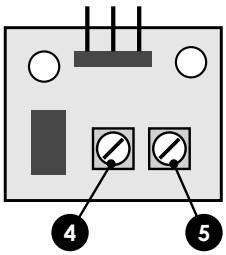
1. .....
2. .....
3. .....
4. .....
5. .....
6. .....
7. .....

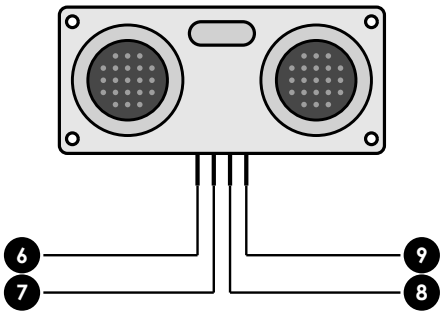
B Label the sensors

9 marks

Write the name of each numbered pin or part.







1. ..... 2. ..... 3. .....

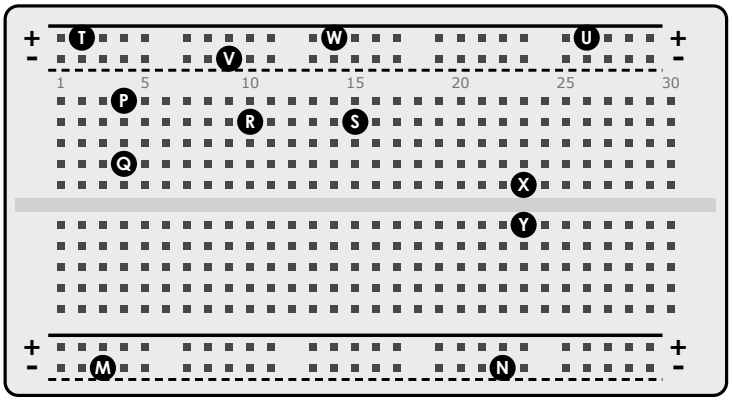
4. ..... 5. ..... 6. .....

7. ..... 8. ..... 9. .....

C Are they connected?

6 marks

Look at the lettered holes of the breadboard.



Holes	Yes or No?
P and **Q**	
R and **S**	
T and **U**	
V and **W**	
X and **Y**	
M and **N**	

D PIR or ultrasonic?

8 marks

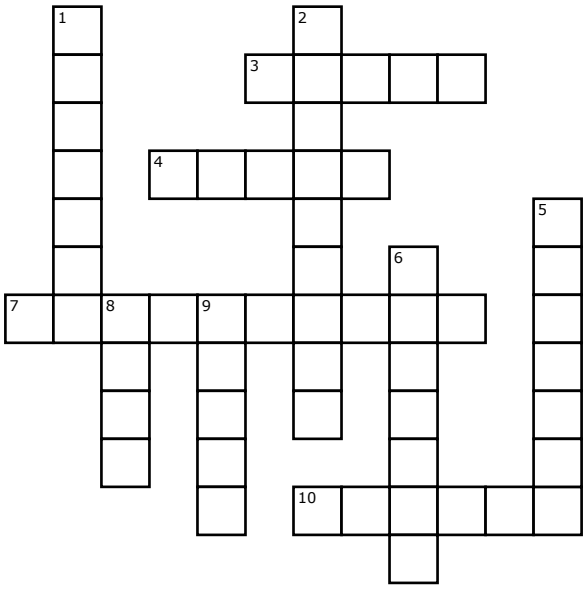
Which sensor is it? Tick one box in each row.

The sensor ...	PIR	Ultrasonic
It estimates the distance from it to an object.	☐	☐
It measures infrared objects, like humans, in its field of view.	☐	☐
It has a TRIGGER pin and an ECHO pin.	☐	☐
Its OUT pin outputs 3.3 V.	☐	☐
It works like a bat finding its route.	☐	☐
Its sensitivity can be adjusted up to 7 m.	☐	☐
It sends ultrasound signals and reads the reflected signals.	☐	☐
It has a trimpot to adjust the time delay of the output.	☐	☐

E Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

3. The system ... synchronizes all the components (5)
4. The company that produces the ATMEGA328p (5)
7. You make circuits on it without welding (10)
10. It detects or measures a physical property (6)

DOWN

1. The input pin of the ultrasonic sensor (7)
2. The micro:bit antenna connects devices using ... (9)
5. This block holds the repetitive work (7)
6. Its Uno board has the ATMEGA328p chip (7)
8. The reflected sound that a bat listens to (4)
9. In a ..., electricity travels only in one direction (5)

REVISION

Unit 4 in One Page

Read this the night before the test

Topic	The key facts
Sensors	A sensor detects or measures a physical property: light, temperature, soil conditions, a person or a distance.
Input, process, output	The sensor collects the data (input), the microcontroller processes it, and a device gives the output, like a bulb that is switched on or off.
Microcontroller	A chip designed to perform some operations according to the given instructions (a program).
Four components	CPU, memory, system clock and peripheral devices (pins).
Memory	**Non-volatile**: keeps its data without electricity and stores the program. **Volatile**: loses its data without electricity; it works as the RAM.
Devices	Solar lamps, washing machines, microwave ovens, traffic lights, computer printers and remote controllers use microcontrollers.
SBC and SCC	Single board computer: one board (Raspberry pi). Single chip computer: one IC (the Arduino chip).
micro:bit front	Buttons A and B, 25 LED bulbs, pins 0, 1, 2, the 3V port and the GND port.
micro:bit back	Bluetooth antenna, CPU, micro USB port, battery connector (3 V), accelerometer and compass, pin edge connector.
Block editor	Drag-and-drop blocks, online. **on start** = the initial action; **forever** = repetitive work. Download the program to the MICRO BIT drive.
Arduino Uno	Mini USB, power regulator, DC power, power pins, analog pins, ATMEGA328p (Atmel), digital pins.
Breadboard, LED	Breadboard: circuits without welding. LED: light in one direction; a + pin (longer leg) and a - pin.
PIR sensor	Finds infrared objects (people). GND, OUT (3.3 V), VCC (5 V); sensitivity (up to 7 m) and delay trimpots.
Ultrasonic sensor	Finds the distance with ultrasound and its echo, like a bat. VCC (5 V), TRIGGER (input), ECHO (output), GND.

TICK WHEN YOU CAN DO IT

- [ ] I can explain input, process and output with the light sensor example.
- [ ] I can name the four components of a microcontroller.
- [ ] I can name devices that use microcontrollers.
- [ ] I can label the front and the back of the micro:bit.
- [ ] I can code the micro:bit with Basic and Input blocks.
- [ ] I can label the Arduino Uno and use a breadboard and an LED.
- [ ] I can explain the PIR sensor and the ultrasonic sensor.

WORD POWER

Unit 4 Word List

Thirty-two words you must know

Word	What it means
Sensor	A device that detects or measures a physical property.
Microcontroller	A chip that performs operations by the given instructions.
Input / Process / Output	Collect data / work on the data / give the information.
CPU	Central Processing Unit: it processes data into information.
Non-volatile memory	Keeps its data without electricity. It stores the program.
Volatile memory	Loses its data without electricity. It works as the RAM.
System clock	An electronic device that synchronizes all the components.
Peripheral devices	Small pins used to input data or output information.
Solar lamp	A lamp with solar cells, light sensors and a microcontroller.
SBC	Single Board Computer: all on one board (Raspberry pi).
SCC	Single Chip Computer: all in one IC (the Arduino chip).
Microcontroller based kit	A board with a microcontroller: micro:bit, Arduino ...
micro:bit	A microcontroller module made by the BBC, with 25 LEDs.
Bluetooth antenna	It connects devices using Bluetooth and sends radio waves.
Accelerometer	An embedded sensor of the micro:bit that measures motion.
Block editor	The drag-and-drop editor for coding the micro:bit online.
on start / forever	Blocks for the initial action / for the repetitive work.
Arduino Uno	The Arduino board used in this unit (chip: ATMEGA328p).
ATMEGA328p	The microcontroller chip of the Arduino Uno, made by Atmel.
Power regulator	It controls the voltage given to the Arduino board.
Breadboard	A board with connected holes, for circuits without welding.
LED	Light Emitting Diode: it gives light when electricity flows.
Diode	A part in which electricity travels only in one direction.
PIR sensor	Passive Infrared sensor: it measures infrared objects.
Trimpot	A knob on the PIR sensor for the sensitivity or the delay.
Ultrasonic sensor	It estimates the distance to an object with ultrasound.
Echo	The reflected sound (or signal) that comes back.
Light sensor	A sensor that detects light: sunrise and sunset.
RAM	Random Access Memory: the volatile memory.
Compass	An embedded sensor of the micro:bit: it finds the direction.
Raspberry pi	A single board computer (a microcontroller based kit).
VCC / GND	The positive terminal / the negative terminal.

"Learn the words today, and the ideas will follow tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 4 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. A microcontroller processes the data collected from
   (a) sensors (b) printers (c) monitors (d) speakers
2. Which memory keeps the program when there is no electricity?
   (a) volatile memory (b) RAM
   (c) non-volatile memory (d) system clock
3. Which of these is a single board computer?
   (a) the Arduino chip (b) Raspberry pi (c) an LED (d) a breadboard
4. How many LED bulbs does the micro:bit have?
   (a) 5 (b) 16 (c) 25 (d) 32
5. Which block holds the work that must be repeated?
   (a) on start (b) forever (c) show string (d) clear screen
6. The block pause (ms) 1000 waits for
   (a) 1 second (b) 10 seconds (c) 100 seconds (d) 1000 seconds
7. The microcontroller chip of the Arduino Uno is the
   (a) ATMEGA328p (b) BBC (c) Bluetooth (d) USB
8. Which sensor works like a bat finding its route?
   (a) PIR (b) ultrasonic (c) light (d) temperature

2 Fill in the blanks

8 marks

One word or number in each blank.

1. The basic functions of a computer are input, process and ..... .
2. The system clock ..... all the components of a microcontroller.
3. Volatile memory functions as the ..... .
4. The micro:bit is connected to a computer with a micro ..... cable.
5. The ..... block holds the initial action of a program.
6. A ..... is used to make circuits without welding.
7. In a diode, electricity travels only in ..... direction.
8. The OUT pin of the PIR sensor outputs ..... V.

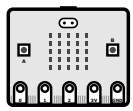
3 Name the pictures

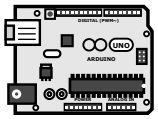
8 marks

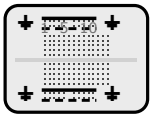
Use the words in the box. Two words are not used.

WORD BANK

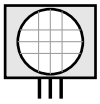
ultrasonic sensor	LED	Raspberry pi	solar lamp	micro:bit
breadboard	washing machine	PIR sensor	microwave oven	Arduino Uno

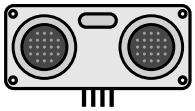
1 

2 

3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

Write in the space given.

1. Name the four basic components of a microcontroller. (4 marks)

   1. ..... 2. .....

   3. ..... 4. .....

2. A solar lamp turns on when it gets dark. Write its input, process and output. (3 marks)

   Input: .....

   Process: .....

   Output: .....

3. Which block would you use: (a) to display a number (b) to erase the displayed data (c) to show a heart? (3 marks)

   (a) ..... (b) ..... (c) .....

4. What is the difference between the on start block and the forever block? (2 marks)

5. Write the four pins of the ultrasonic sensor. Which pin gives the output? (2 marks)

   Pins: ..... Output pin: .....

6. What is a single chip computer? Give one example. (2 marks)